Python Engineering

WEEK 02 - Data Structures | CLASS 07 - NumPy Arrays | EX15

Official class change: class_06_dictionaries_sets is complete. We enter **class_07_numpy_arrays**, still inside week_02_data_structures.

EX15 - Trajectory analysis with NumPy

So far trajectory is a list of tuples. NumPy lets us convert that structure into a **numerical matrix** and operate on complete columns without writing an explicit loop for every calculation.

```
import numpy as np

trajectory_array = np.array(trajectory)

print(trajectory_array.shape)
# (N, 3)
```

1. Shape: rows and columns

We keep your `(h, v, t)` convention. Each row is one state and each column is one variable.

```
heights    = trajectory_array[:, 0]
velocities = trajectory_array[:, 1]
times      = trajectory_array[:, 2]
```

`:` means 'all rows'. Thus `trajectory_array[:, 0]` means all rows from column 0.

2. NumPy indexing

Expression	Meaning
A[0, 0]	row 0, column 0
A[0, :]	complete first row
A[:, 0]	complete first column
A[-1, :]	complete last row
A[2:5, 1]	rows 2..4 from column 1

3. Key difference: list vs ndarray

```
python_list = [1, 2, 3]
python_list * 2
# [1, 2, 3, 1, 2, 3]

array = np.array([1, 2, 3])
array * 2
# [2, 4, 6]
```

NumPy arithmetic is applied element by element. This is one of the main reasons NumPy is the foundation of scientific Python.

4. Vectorization

```
specific_energy = config['g'] * heights + 0.5 * velocities**2
```

The equation is applied to every trajectory point at once. No explicit for loop is needed for each value.

5. Maximum and maximum location

```
max_h = np.max(heights)
max_index = np.argmax(heights)
max_state = trajectory_array[max_index, :]
```

```
h_max, v_at_max, t_at_max = max_state
```

`np.max()` gives the maximum value. `np.argmax()` gives the index where it occurs.

6. Your mission

```
Create: ex15_numpy_trajectory_analysis.py

1. Reuse your current simulator and obtain trajectory.
2. Convert trajectory to np.array.
3. Print shape and ndim.
4. Extract h, v, t as columns.
5. Print first and last state using NumPy indexing.
6. Compute max altitude using np.max().
7. Find its index with np.argmax().
8. Recover h, v, t at the maximum state.
9. Compute specific mechanical energy for every state.
10. Compare list * 2 with np.array * 2 using a tiny example.
```

7. Checks I want to see

```
trajectory_array.shape == (len(trajectory), 3)
trajectory_array[0, :] matches initial state
trajectory_array[-1, :] matches final state
np.max(heights) matches previous max_h
v_at_max should be close to 0 m/s
```

8. Not yet

Do not use pandas, do not plot, and do not rewrite Euler with NumPy yet. First we want solid control of arrays, shape, indexing, slicing, and vectorized operations.

Learning Tracker

Concept	EX15
np.array()	New
shape / ndim	New
2D indexing	New
row/column slicing	New
element-wise arithmetic	New
vectorization	New
np.max() / np.argmax()	New